Xinbo Zhang, Luya Wang and Baosheng Chen*

Recombinant HDL (Milano) protects endotoxin-challenged rats from multiple organ injury and dysfunction

Abstract: Endotoxemia, the systemic inflammatory host response to infection, leads to severe septic shock and multiple organ injury and dysfunction syndrome (MOPS), which cause mortality. Apolipoprotein A-I_{Milano} ($apoAI_M$), a naturally occurring cysteine mutant of apoAI with dimers as its effective form, showed an enhanced cardiovascular protective activity compared with wild-type apoAI (apoAIwt). To investigate the role of recombinant high-density lipoprotein (rHDL) reconstituted with $apoAI_M$ ($rHDL_M$) on endotoxemia and MOPS, we examined the anti-inflammatory, anti-oxidant, and protective effects of this cysteine mutant against organ injury in endotoxin-challenged rat models compared with rHDLwt. In the present study, we demonstrated for the first time that pretreatment with $rHDL_M$ significantly attenuated liver and renal dysfunction and histopathological features of lung injury in endotoxin-challenged endotoxemia rats. Administration of $rHDL_M$ to endotoxemia rats dramatically suppressed proinflammatory cytokines and adhesion molecule increase in tumor necrosis factor α, interleukin 1β, interleukin 6, and intercellular adhesion molecule 1. In addition, $rHDL_M$ pretreatment inhibited lipid peroxidation and enhanced total antioxidant capacity *in vivo*. In comparison with rHDLwt, $rHDL_M$ showed enhanced capacity on anti-inflammatory and anti-oxidant functions. In summary, administration of $rHDL_M$ protected endotoxin-challenged endotoxemia and MOPS through enhanced anti-inflammatory and anti-oxidant properties.

Keywords: apolipoprotein A-I; cysteine mutant; endotoxemia; multiple organ dysfunction.

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Introduction

Sepsis or endotoxemia is defined as the systemic inflammatory host response to infection and a serious clinical problem that causes mortality (Wheeler and Bernard, 1999; Angus et al., 2001). In patients and animals with sepsis, endotoxemia caused by endotoxin activates inflammatory responses, which can lead to severe septic shock and multiple organ injury and dysfunction syndrome (MOPS). Endotoxin, a component of the outer wall of Gram-negative bacteria, accesses blood circulation and initiates the production of biologically active molecules including proinflammatory cytokines such as tumor necrosis factor α (TNFα), interleukin 1β (IL-1β), interleukin 6 (IL-6), and reactive oxygen species (ROS) during the development of endotoxemia (Bhattacharyya et al., 2004). Owing to increased production of proinflammatory cytokines and ROS, endotoxin induces extensive damage to a variety of organs, including the liver, kidneys, and lungs.

High-density lipoprotein (HDL), the smallest lipoprotein in circulation, is inversely correlated with the incidence of coronary artery disease, which is attributed to its role in reverse cholesterol transport (RCT) and anti-inflammatory and anti-oxidant properties (Tall, 1990; Ansell et al., 2004; Luscher et al., 2014). Apolipoprotein A-I (apoAI) is the active protein component responsible for RCT process in HDL. Recombinant HDL (rHDLwt) reconstituted with wild-type apoAI (apoAIwt) can inhibit endotoxin-induced expression of proinflammatory cytokines and adhesion molecules (Hubsch et al., 1995; Yan et al.,

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2006) and improve the subsequent multiple organ dysfunction (McDonald et al., 2003). Furthermore, administration of rHDLwt improved human endotoxemia *in vivo* by binding and neutralizing lipopolysaccharide (LPS)/endotoxin and by reducing CD14 expression on monocytes (Pajkrt et al., 1996). ApoAI$_{Milano}$ (apoAI$_M$), a natural cysteine mutant of apoAI with dimers as its effective form, has been shown to enhance cardiovascular-protective activities (Franceschini et al., 1988; Shah et al., 1998; Ibanez et al., 2008; Speidl et al., 2010). Previously, our laboratory showed that recombinant HDL reconstituted with apoAI$_M$ (rHDL$_M$) improved the anti-inflammatory and anti-oxidant properties and enhanced protective activity in atherosclerosis (Zhang et al., 2010). However, the role of rHDL$_M$ on endotoxemia and MOPS still need further evaluation to provide definitive evidence.

In this study, we are able to show that pretreatment with rHDL$_M$, in comparison with rHDLwt, showed improved protection on the liver and renal dysfunction and histopathological features of lung injury through enhanced anti-inflammatory and anti-oxidant properties in endotoxin-challenged rat model.

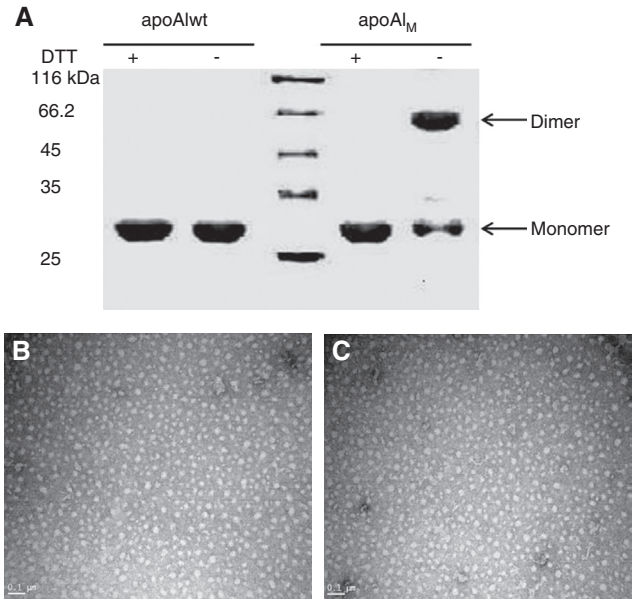


Figure 1 Characteristics of reconstituted apoAI and recombinant HDL.
(A) SDS-PAGE of reconstituted apoAI in the absence or presence of DTT. Reconstituted apoAI$_M$ exists as both a dimer and a monomer, whereas apoAIwt existed as monomer. Electron micrographs of the negatively stained rHDLwt (B) and rHDL$_M$ (C) nanoparticles were observed using a transmission electron microscope (100 000× original magnification).

Results

Characteristics of reconstituted apoAI and rHDLs

Purified apoAIwt and apoAI$_M$ were detected with SDS-polyacrylamide gel electrophoresis (SDS-PAGE) in the absence or presence of dithiothreitol (DTT), an unusually strong reducing agent (Figure 1A). Reconstituted apoAIwt exists as a monomer, whereas reconstituted apoAI$_M$ exists as both a dimer and a monomer, in accordance with a previous report (Weisgraber et al., 1983). HDL is the smallest of the lipoproteins. Based on morphological observation with transmission electron microscope, we confirmed that the micellar complexes of reconstituted apoAIwt or apoAI$_M$ with soybean lecithin (SPC) were discoidal particles ranging from 20 to 35 nm. The average size of rHDLwt and rHDL$_M$ were 26.38±4.87 and 27.77±5.17 nm, respectively.

Effects of rHDL$_M$ on serum lipid profiles

The effect of rHDL administration on serum lipid levels 6 h after endotoxin treatment is shown in Table 1. In comparison with the endotoxin group, rHDLwt and rHDL$_M$ injection increased serum HDL-cholesterol (HDL-C), whereas SPC injection had no influence. Recombinant HDL injection including rHDLwt and rHDL$_M$ significantly increased serum total cholesterol (TC) level, which is mainly attributed to increased HDL-C (Table 1).

Effects of rHDL$_M$ on the multiple organ dysfunction caused by endotoxin

When compared with rats treated with saline vehicle, endotoxin injection resulted in significant increases

Table 1 Serum lipid profiles 6 h after endotoxin injection.

Group	TC	TG	HDL-C	Non-HDL-C
Saline	2.05±0.24	1.39±0.05	1.21±0.07	0.46±0.33
Endotoxin	1.71±0.16[a]	1.43±0.74	0.90±0.06[a]	0.58±0.07
SPC	1.79±0.10	1.44±0.28	0.88±0.10[a]	0.54±0.13
rHDLwt	2.22±0.24[b]	1.40±0.26	1.09±0.16[b]	0.67±0.11
rHDL$_M$	2.19±0.31[b]	1.66±0.40	1.09±0.19	0.63±0.05

Results expressed as mean±SD.
[a]$p<0.05$, compared with the saline group.
[b]$p<0.05$, compared with the SPC group.

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in the serum levels of alanine aminotransferase (ALT), aspartate aminotransferase (AST) (liver injury and dysfunction, Figure 2A and B), urea, creatinine (renal dysfunction, Figure 2C and D), and creatine kinase (CK, heart and skeletal muscle injury, Figure 2E). Pretreatment of endotoxin-challenged rats with rHDLs caused significant reduction in the serum levels of ALT and AST, whereas SPC pretreatment only has slight reduction (Figure 2A and B). In addition, treatment with rHDL$_M$ resulted in further reduction of serum ALT and AST than rHDLwt injection (Figure 2A and B). Pretreatment of endotoxemia rats with rHDLwt and rHDL$_M$ also significantly improved the serum level of urea and creatinine, which reflects the improvement of renal dysfunction (Figure 2C and D). However, pretreatment with rHDLs did not affect the increase in serum CK level but had a decreasing trend (Figure 2E).

Effects of rHDL$_M$ on the lung injury caused by endotoxin

Figure 3 illustrates the hispathological micrographs of a normal lung without endotoxin administration (Figure 3A) and histological signs of lung injury following endotoxin treatment including pulmonary edema, infiltration of inflammatory cells, and destruction of the normal architecture of the lungs (Figure 3B). Pretreatment of endotoxemia rats with rHDLwt and rHDL$_M$ attenuated these histological changes associated with endotoxin administration (Figure 3D and E).

Effects of rHDL$_M$ on serum inflammatory cytokines and chemokine level

When compared with saline control rats, administration of endotoxin resulted in enhanced expression of serum inflammatory cytokines (TNFα, IL-1β, and IL-6) and adhesion molecules [intercellular adhesion molecule 1 (ICAM-1)] (Figure 4A). Pretreatment of endotoxemia rats with rHDLwt attenuated the expression of serum inflammatory cytokines and ICAM-1 caused by endotoxin (Figure 4A). Furthermore, pretreatment with rHDL$_M$ resulted in further improvement of serum cytokines and chemokine level compared with rHDLwt injection (Figure 4A).

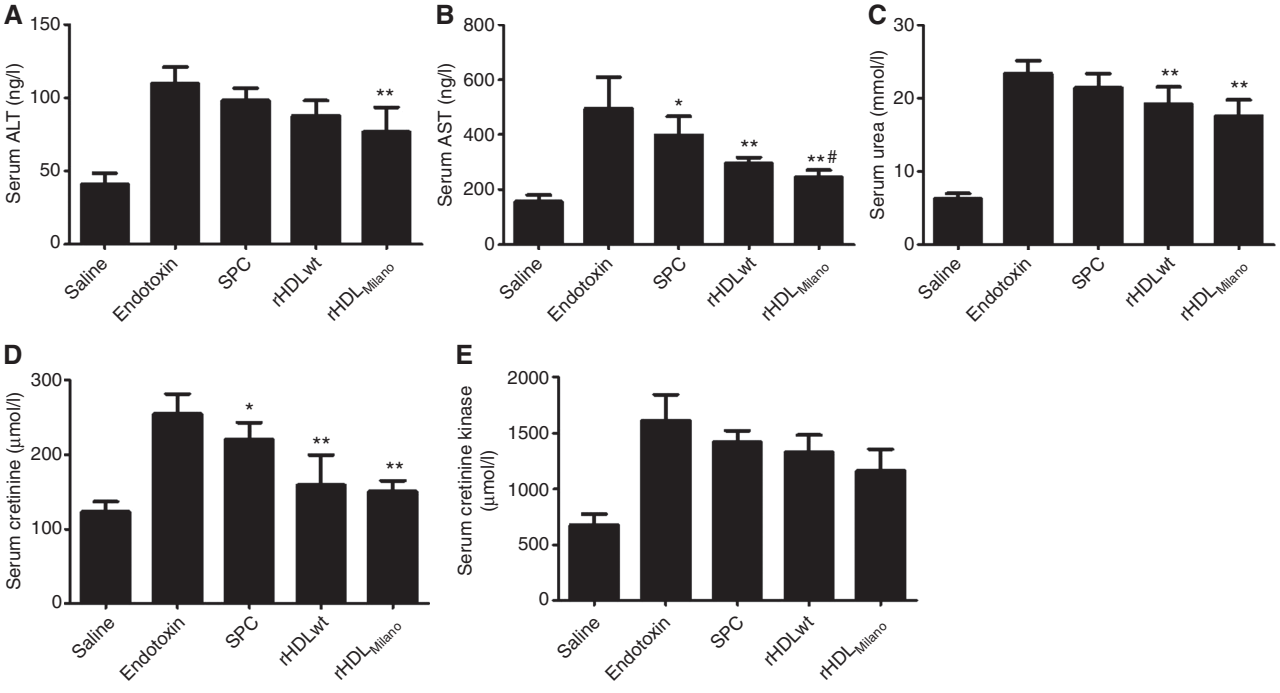


Figure 2 Effect of rHDL$_M$ and rHDLwt on endotoxin-induced liver and renal injury.
Pretreatment of endotoxin-challenged rats with saline, SPC, or rHDLs 15 min prior to a single injection of endotoxin (400EU/kg, iv) or saline. Six hours after endotoxin injection, serum ALT (A) and AST (B), urea (C), and cretinine (D) levels were measured as described in Materials and methods. Results shown are means±SD. *p<0.05 and **p<0.01, compared with the endotoxin group; $^{\#}p$<0.05, compared with the rHDLwt group.

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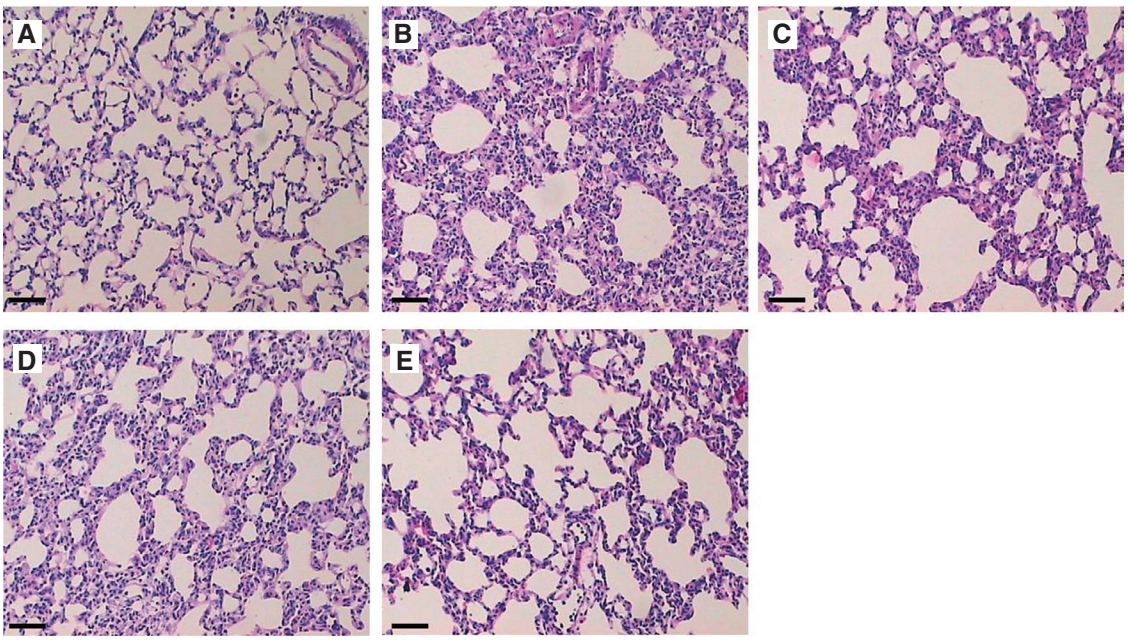


Figure 3 Morphological changes in rat lung sections of rHDL$_M$ and rHDLwt on endotoxin-challenged endotoxemia.
(A) Saline, (B) endotoxin, (C) SPC, (D) rHDLwt, and (E) rHDL$_M$ groups (H&E staining, 100× original magnification, scale bar: 100 μm).

Effects of rHDL$_M$ on serum lipid peroxidation and antioxidant activity

Endotoxin injection resulted in lipid peroxide formation, causing decreased levels of scavengers or quenchers of free radicals (Sakaguchi and Furusawa, 2006). When compared with saline control rats, administration of endotoxin resulted in significantly increased expression of malondialdehyde (MDA, Figure 4B), a specific marker of lipid peroxidation, and serum nitric oxide (NO) level (Figure 4C). Pretreatment of endotoxemia rats with rHDL$_M$ significantly attenuated serum MDA and NO expression, whereas rHDLwt treatment only showed slight decrease of these markers (Figure 4B and C). In accordance with serum lipid peroxidation, administration of rHDLwt and rHDL$_M$ led to an increased serum total antioxidant capacity (T-AOC) (Figure 4D).

Discussion

Endotoxin-induced endotoxemia was characterized by infiltration of inflammatory cells in the liver, lungs, and kidneys; increased production of ALT, AST, urea, creatinine, and CK, indicating liver, renal, heart, and skeletal muscle injury and dysfunction; and morphological evidence of tissue injury. Here, we demonstrate for the first time that pretreatment with rHDL$_M$ reconstituted with recombinant apoAI$_M$ significantly attenuated liver and renal dysfunction and lung injury in endotoxin-challenged endotoxemia rats. The administration of rHDL$_M$ to endotoxemia rats significantly decreased the expression of proinflammatory cytokines TNFα, IL-1β, and IL-6 and adhesion molecule ICAM-1. In addition, rHDL$_M$ treatment can inhibit lipid peroxidation and enhance T-AOC. In comparison with rHDLwt, rHDL$_M$ showed enhanced capacity on anti-inflammatory and anti-oxidant functions. The improvement of rHDL$_M$ on endotoxemia and MOPS may be attributed to its enhanced anti-inflammatory and anti-oxidant properties.

In response to endotoxic shock, systemic release of proinflammatory cytokines (TNF-α, IL-6, and IL-1β) results in tissue damage and organ dysfunction (Haveman et al., 1999). Inhibition of the proinflammatory cytokines may exert beneficial effects on endotoxemia (Haveman et al., 1999). HDL and rHDL are well known for neutralizing LPS/endotoxin, decreasing inflammatory cytokines production, and inhibiting the expression of cell adhesion molecules *in vitro* and *in vivo* (Parker et al., 1995; Cockerill et al., 2001). In our study, we observed a greater anti-inflammatory effect of rHDL$_M$ than rHDLwt in endotoxin-challenged endotoxemia rat model, which is in accordance with previous reports (Zhang et al., 2010; Ibanez et al., 2012). Previous observations suggested that apoAI, the major protein component in HDL, possesses intrinsic anti-inflammatory

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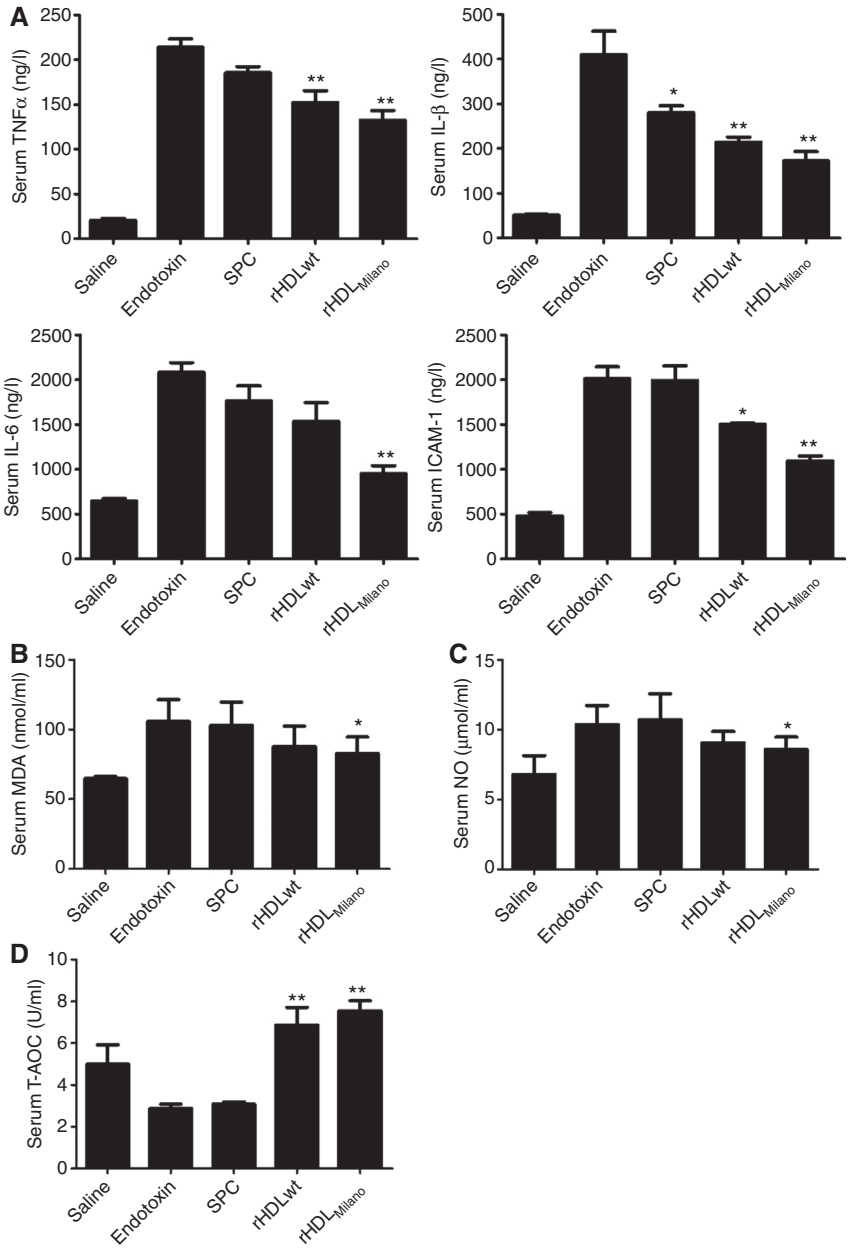


Figure 4 Effect of rHDL$_M$ and rHDLwt on serum proinflammatory cytokines, adhesion molecule, and oxidative stress following endotoxin-induced endotoxemia.

(A) Pretreatment of endotoxin-challenged rats with saline, SPC, or rHDLs 15 min prior to a single injection of endotoxin (400EU/kg, iv) or saline. Six hours after endotoxin injection, serum TNFα, IL-1β, IL-6, and ICAM-1 levels were detected with ELISA. (B) Administration of rHDL$_M$ led to increased inhibition of MDA (an end product of lipid peroxidation) and NO production and increased serum T-AOC in comparison with rHDLwt. Results shown are means±SD. *p<0.05 and **p<0.01, compared with the endotoxin group.

and anti-oxidant properties of HDL (Moore et al., 2005). In addition, the phospholipid composition of rHDL also exerts inhibitory activity on the expression of inflammatory cytokines (Baker et al., 2000). Furthermore, the phospholipid composition of HDL can bind bacteria LPS/endotoxin directly or with the help of LPS-binding protein to inhibit LPS-induced inflammatory response (Wurfel et al., 1994; Parker et al., 1995). We reported previously that reconstituted apoAI$_M$ exhibited similar lipid binding ability with apoAIwt through dimyristoylphosphatidylcholine (DMPC) clearance assay (Zhu et al., 2005), suggesting that the enhanced anti-inflammatory effects of rHDL$_M$ may depend on specific properties of apoAI$_M$ protein itself.

ROS and reactive nitrogen species (RNS, e.g., NO$^{\cdot}$) are well known for their dual role as both deleterious and beneficial species (Valko et al., 2007). Overproduction

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of ROS or RNS results in oxidative stress leading to lipid peroxidation and damage of cell structures. In normal physiological conditions, many antioxidants including glutathione, vitamin C, and superoxide dismutase protect against oxidative stress to maintain the 'redox homeostasis'. Endotoxin is known to cause oxidative stress *in vivo*. Our findings indicated that the administration of rHDL$_M$ exerts better effects on inhibiting oxidative stress, as reflected by the levels of NO and MDA, the end products of lipid peroxidation, and enhancing T-AOC. Consistent with our findings, Ibanez et al. (2012) also observed decreased MDA production in the rHDL$_M$ group compared with the rHDLwt group. The antioxidant activity of HDL *in vivo* is partly attributed to the paraoxonase 1 (PON-1) and platelet-activating factor acetylhydrolase (PAF-AH), two antioxidant-like enzymes associated with natural HDL. It has been reported that cysteine mutations of apoAI$_M$ significantly increased PON activity relative to wild type (Oda et al., 2001) and protected phospholipids from free radical-mediated lipid peroxidation at an earlier step in the oxidation cascade (Bielicki and Oda, 2002). In addition, a cross-talk existed between proinflammatory cascade and oxidative stress. The inhibition of rHDL$_M$ on endotoxin-induced inflammatory response and cytokine release may decrease ROS production and systemic oxidative stress.

In conclusion, we have shown that administration of rHDL$_M$ protected endotoxin-challenged endotoxemia and MOPS through enhanced anti-inflammatory and anti-oxidant properties.

Materials and methods

Preparation of recombinant HDL

The expression, purification, and endotoxin removal of recombinant proteins, including apoAIwt and apoAI$_M$, were performed as described previously (Zhu et al., 2005; Zhang et al., 2010). The endotoxin level of reconstituted apolipoproteins was checked with tachypleus amebocyte lysate before injection to preclude endotoxic contamination. Recombinant HDLs were prepared *via* sodium cholate dialysis with SPC (at least 90% purification, Huaqing Mayheng Natural Product technology Development Co., Ltd, Beijing, China) with a mass ratio of 3.35:1 (SPC:apoAI) and sterilized by filtering through a 0.22-μm membrane (Zhang and Chen, 2010; Zhang et al., 2010).

Size and morphology of rHDLs

Recombinant HDLs reconstituted with apoAIwt and apoAI$_M$ were dialyzed against 0.01 M phosphate-buffered saline (PBS, pH 7.4) and diluted to appropriate concentration (200 μg/ml) to determine size and morphology with electron microscope (JEM-1010, Japan Electron Optics Laboratories, Tokyo, Japan) as described previously (Zhang and Chen, 2010). Images of rHDLs were obtained and analyzed with at least 200 particles using Image Proplus 6.0 software (Media Cybernetics, Silver Spring, USA) to detect the mean diameter.

Induction of endotoxin-challenged sepsis and tissue injury

Male Wister rats (SPF grade, 12 weeks old, 190–240 g) were obtained from the Department of Laboratory Animal Science of Peking University Health Science Center. Standard laboratory chow and drinking water were provided *ad libitum*. All animal experimental work was approved by the animal care committee of the Chinese Academy of Medical Science. Endotoxin-challenged endotoxemia was induced as described previously (Hasegawa et al., 1996). Thirty rats were randomly divided into five groups as described in Table 1: group 1, saline control (saline, n=6); group 2, endotoxin control (endotoxin, 400 EU/kg Gram-negative bacterial endotoxin, n=6); group 3, SPC (135 mg/kg, n=6); group 4, rHDLwt (40 mg/kg, n=6); group 5, rHDL$_M$ (40 mg/kg, n=6). Fifteen minutes before endotoxin injection, normal saline, SPC, or rHDLs were injected, respectively, to the rats of each group through the dorsal artery of the penis. Subsequently, saline was injected into saline animals, and endotoxin was injected into the endotoxin, SPC, rHDLwt, and rHDL$_M$ groups through the dorsal artery of the penis (Supplementary Table 1).

Serum lipid profiles and cytokines analysis

Six hours after endotoxin injection, blood samples were obtained from the retro-orbital plexus. Serum was isolated and stored at -80°C until analysis. Serum TC and triglyceride (TG) concentrations were determined by standard enzymatic methods. Serum HDL levels were measured using a direct HDL-C reagent kit (BioSino Bio-technology and Science, Beijing, China). Non-HDL-C was determined as the difference between serum TC and HDL-C. Serum TNFα, IL-1β, and IL-6, markers of inflammation, and serum ICAM-1 level were determined *via* enzyme-linked immunosorbent assay (ELISA) (Boster Biological Technology, Wuhan, China).

Measurement of serum lipid peroxidation levels, serum NO, and T-AOC assay

Serum MDA, the end product of lipid peroxidation, was analyzed by a spectrophotometric method described previously (Garca et al., 2013). Serum NO and T-AOC level were determined *via* commercially available kits from Nanjing Jiancheng Bioengineering Institute, China.

Quantification of organ function and injury

Liver injury and dysfunction were evaluated by measuring the increase in serum levels of ALT and AST. Renal dysfunction and injury was assessed by measuring the increases in serum concentrations of creatinine and urea (indicators of renal failure). Cardiac

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and skeletal muscle damage was assessed by the serum level of CK. All these injury and dysfunction markers were analyzed by commercially available kits from BioSino Bio-technology and Science.

Histopathological analysis

The lungs were harvested 6 h after endotoxin injection. The thoracic cavity was exposed immediately, and *in situ* perfusion fixation through the left cardiac ventricle was performed by a thorough perfusion with PBS for 10 min, followed by 10% neutral buffered formaldehyde solution for 30 min (Zhang et al., 2010). The samples were fixed in 10% buffered formaldehyde solution at room temperature, dehydrated by graded ethanol, and embedded in Paraplast. Sections (thickness 5 μm) were deparaffinized with xylene, stained with hematoxylin and eosin (H&E), and examined using light microscopy (Leica DMI 4000 B, Leica Microsystems, Mannheim, Germany).

Statistics

Data were expressed as mean±SD. Differences among the experimental groups were examined for significance using one-way ANOVA followed by multiple comparisons test. $p<0.05$ was considered statistically significant.

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